SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**
Question Count: 4

Code of Conduct

I KENISHA JUDITH · J (192572071) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: J-Kenisha

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

CLOUD COMPUTING FUNDAMENTALS

DEFINITION

- Internet-based computing
- shared resources
- Pay-as-you use
- on-demand services

CHARACTERISTICS

- On-demand access
- Resource pooling
- Broad network access
- Rapid elasticity
- Measured service

BENEFITS

- cost saving
- scalability
- Flexibility
- Accessibility
- High availability

LIMITATIONS

- Internet dependency
- security risks
- Privacy concerns
- Downtime
- Vendor lock-in
- Limited control

Explanation :-

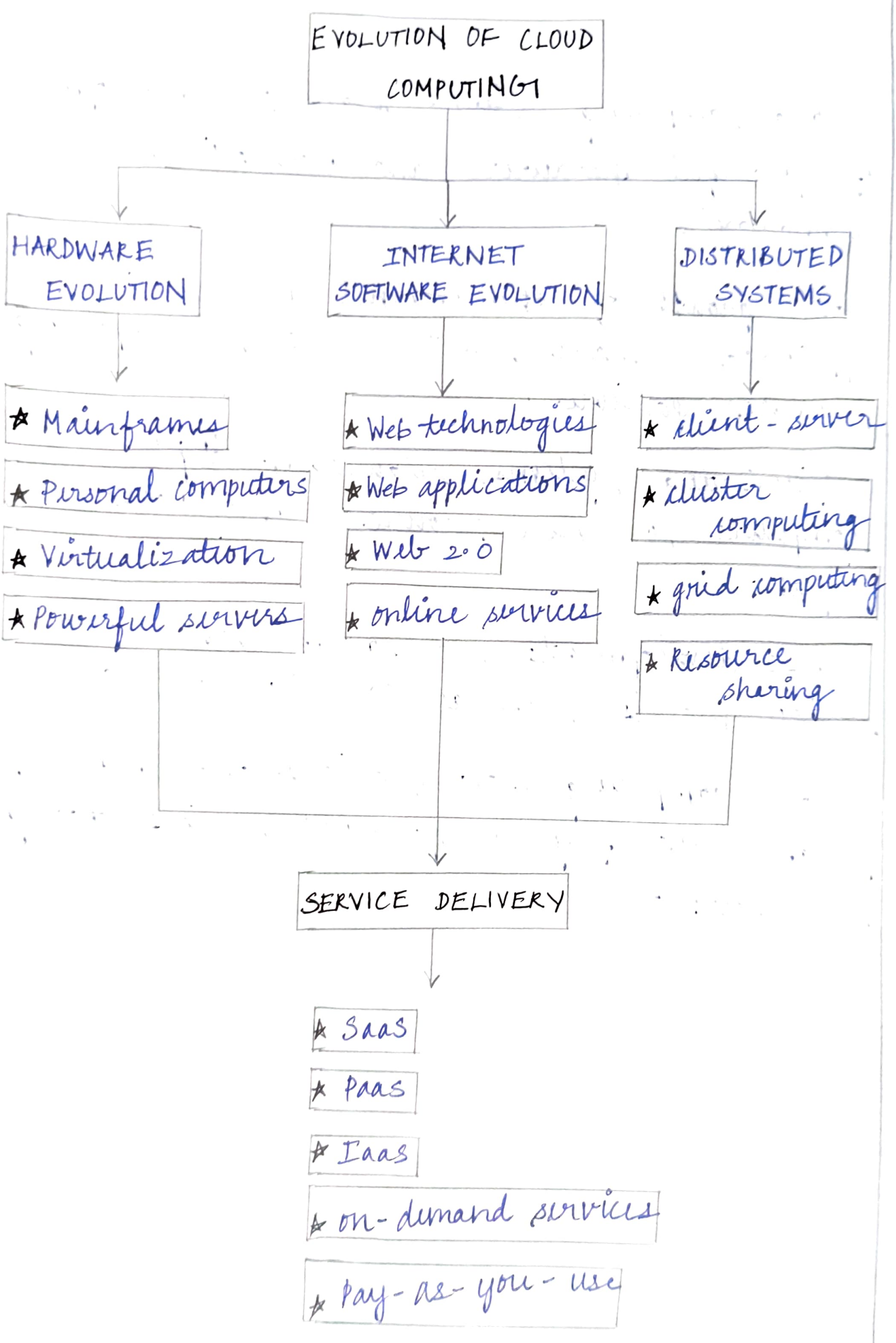
* cloud computing provides computing resources such as storage, servers, applications and databases through the Internet.

* It offers important characteristics such as on-demand access, resource pooling, rapid scalability and measured usage.

* cloud computing provides several benefits, including cost reduction, flexibility, accessibility, scalability, and high availability.

* It allows users and organizations to access resources from anywhere without maintaining large physical infrastructure.

* However cloud computing has some limitations such as Internet dependency, security and privacy concerns, downtime and vendor lock-in.


EVOLUTION OF CLOUD COMPUTING
HARDWARE EVOLUTION
INTERNET SOFTWARE EVOLUTION
DISTRIBUTED SYSTEMS
Mainframes
Personal computers
Virtualization
Powerful servers
Web technologies
Web applications
Web 2.0
online services
client - server
cluster computing
grid computing
Resource sharing
SERVICE DELIVERY
SaaS
PaaS
IaaS
on-demand services
Pay-as-you-use

Explanation :-

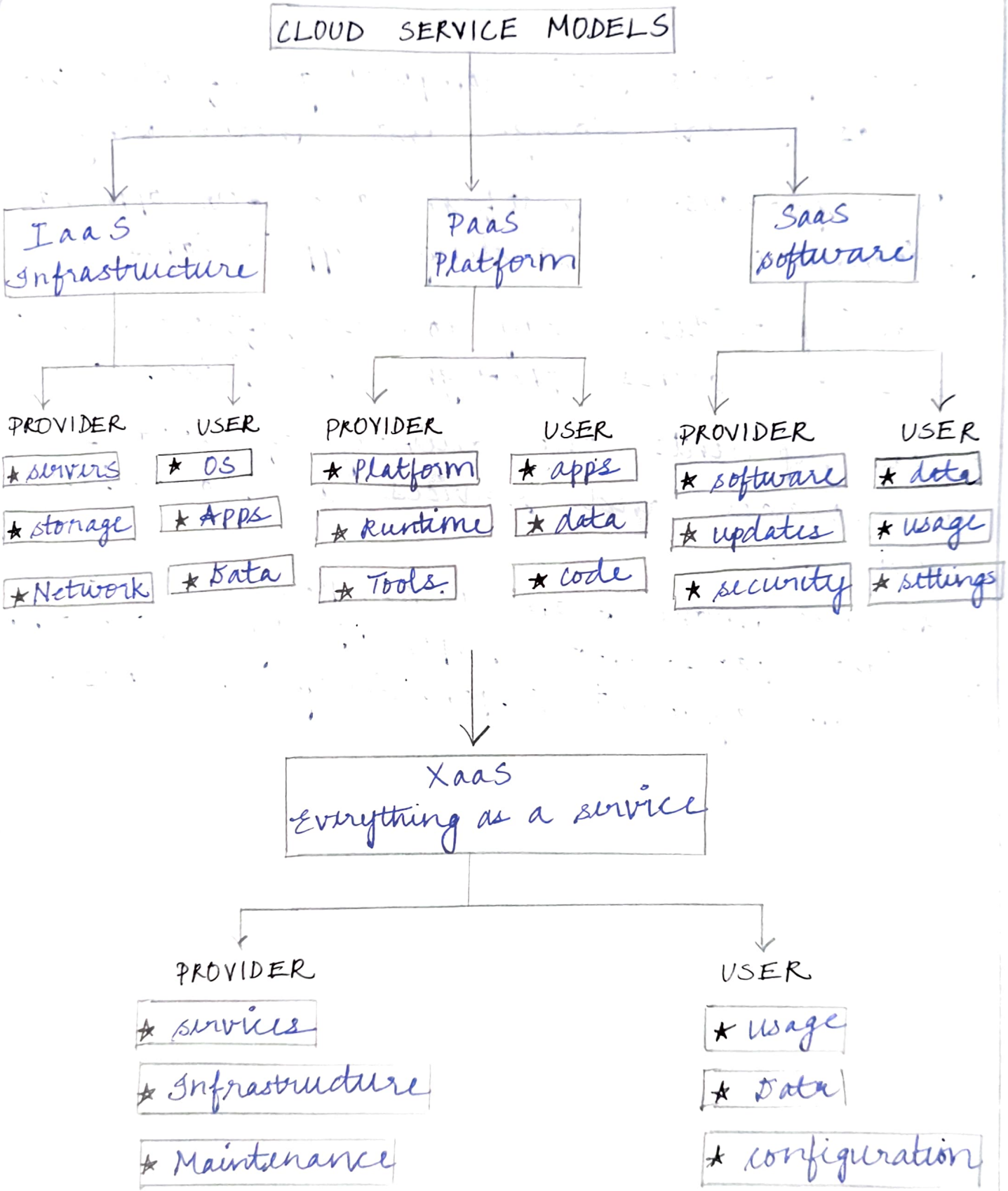
* cloud computing evolved from the development of powerful hardware and virtualization technologies.

* The growth of the Internet and web applications made remote access to computing resources possible.

* Distributed systems allowed multiple computers to share resources and work together.

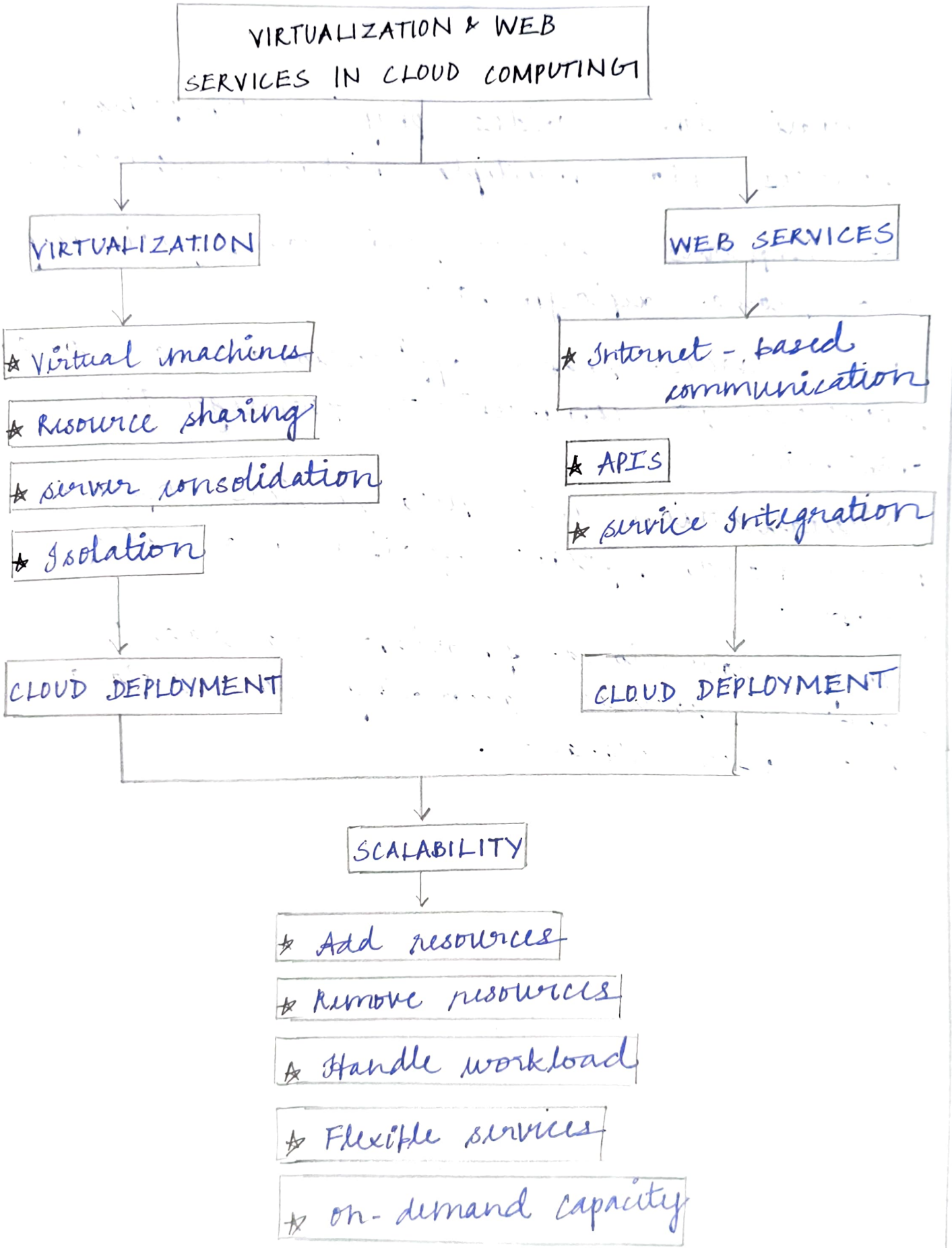
* These developments led to modern cloud service models such as Iaas, Paas and Saas.

* cloud computing now provides on-demand, scalable and pay-as-you use services through the Internet.


CLOUD SERVICE MODELS
IaaS
Infrastructure
PaaS
Platform
Saas
software
PROVIDER
USER
PROVIDER
USER
PROVIDER
USER
★ servers
★ storage
★ Network
★ OS
★ Apps
★ Data
★ Platform
★ Runtime
★ Tools
★ apps
★ data
★ code
★ software
★ updates
★ security
★ data
★ usage
★ settings
XaaS
Everything as a service
PROVIDER
USER
★ services
★ Infrastructure
★ Maintenance
★ usage
★ Data
★ configuration

Explanation :

* Iaas provides basic computing infrastructure such as servers, storage and networking.
* Paas provides a platform and development environment for creating applications.
* Saas provides ready-to-use software that users can access through the internet.
* Xaas represents the delivery of various IT resources and services as cloud-based services.
* As we move from Iaas → Paas → Saas, the provider takes more responsibility while the user's management responsibility decreases.


VIRTUALIZATION & WEB SERVICES IN CLOUD COMPUTING
VIRTUALIZATION
WEB SERVICES
Virtual machines
Resource sharing
server consolidation
Isolation
Internet - based communication
APIs
service Integration
CLOUD DEPLOYMENT
CLOUD DEPLOYMENT
SCALABILITY
Add resources
Remove resources
Handle workload
Flexible services
on-demand capacity

Explanation :

* Virtualization divides physical computing resources into multiple virtual resources.
* It improves resource utilization, flexibility and cloud deployment.
* Web services enable applications and cloud resources to communicate through the Internet.
* Virtualization and web services together make cloud systems flexible and easily scalable.
* They allow cloud providers to add or remove resources according to workload and user demand.